

=== CONFIGURACION GNB (1779964141) ===

This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 20 MHz bandwidth
To run the srsRAN Project gNB with this config, use the following command:
sudo ./gnb -c gnb.yml

```
cu_cp:
  amf:
    addr: 192.168.70.5          # The address or hostname of the AMF (Core 5G).
    port: 38412
    bind_addr: 192.168.70.10    # A local IP that the gNB binds to for traffic from the AMF.
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "00101"
            tai_slice_support_list:
              - sst: 1          # SST=1 for eMBB use case (LAN Party).
    inactivity_timer: 7200      # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200]

ru_sdr:
  device_driver: zmq           # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=23.04e6 # Optionally pass argument
  srate: 23.04                # RF sample rate might need to be adjusted according to selected bandwidth.
  tx_gain: 75                  # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75                  # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 620000            # ARFCN of the downlink carrier (center frequency). For Band n78, 620000 corresponds to 3.5 GHz.
  band: 78                    # The NR band (n78).
  channel_bandwidth_MHz: 20   # Bandwidth in MHz. Number of PRBs will be automatically derived.
  common_scs: 30              # Subcarrier spacing in kHz used for data.
  plmn: "00101"              # PLMN broadcasted by the gNB.
  tac: 7                      # Tracking area code (needs to match the core configuration).
  pdccch:
    common:
      ss0_index: 0            # Set search space zero index to match srsUE capabilities
      coreset0_index: 12     # Set search CORESET Zero index to match srsUE capabilities
    dedicated:
      ss2_type: common        # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1     # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64          # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64          # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log      # Path of the log file.
  all_level: info             # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false           # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
  ngap_enable: false          # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true           # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

metrics:
  layers:
    enable_rlc: true          # Enable RLC metrics
```

```
enable_sched: true          # Enable DU scheduler metrics
periodicity:
  du_report_period: 1000    # Set DU statistics report period to 1s
```

=== CONFIGURACION UE (1779964141) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 23.04e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=23.04e6

[rat.eutra]
dl_earfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78 # Matches gNB Band n78
nof_carriers = 1

max_nof_prb = 106 # For 20 MHz bandwidth, 30 kHz SCS
nof_prb = 106 # For 20 MHz bandwidth, 30 kHz SCS
dl_nr_arfcn = 620000 # Matches gNB ARFCN
ssb_nr_arfcn = 619990 # SSB ARFCN, typically close to DL ARFCN

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 001010123456780 # IMSI with PLMN 00101

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0
ip_addr = 192.168.70.11 # UE tunnel interface IP

[gui]
enable = false
```

=== DOCKER COMPOSE (1779964141) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  env_file:
    - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
  privileged: true
  ports:
    - "9898:9999/tcp"
    - "38412:38412/sctp" # AMF N2 port
    - "2152:2152/udp"   # UPF N3 port
  command: 5gc -c open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: 192.168.70.5 # Core 5G IP
```

```
gnb:
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
  # privileged mode is required only for accessing usb devices
  privileged: true
  # Extra capabilities always required
  cap_add:
    - SYS_NICE
    - CAP_SYS_PTRACE
  volumes:
    # Access USB to use some SDRs
    - /dev/bus/usb/:/dev/bus/usb/
```

```

# Sharing images between the host and the pod.
# It's also possible to download the images inside the pod
- /usr/share/uhd/images:/usr/share/uhd/images
# Save logs and more into gnb-storage
- gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's the easiest
networks:
  ran:
    ipv4_address: 192.168.70.10 # gNB IP
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

ue:
  container_name: srsran_ue
  image: srsran/ue # Assuming srsUE image is available
  privileged: true
  cap_add:
    - NET_ADMIN # Required for creating tun device
  volumes:
    - ue-storage:/tmp
  configs:
    - ue_config.conf
  networks:
    ran:
      ipv4_address: 192.168.70.11 # UE IP
  depends_on:
    gnb:
      condition: service_healthy # UE depends on gNB
  command: ue -c /ue_config.conf

configs:
  gnb_config.yml:
    file: ${GNB_CONFIG_PATH:-./configs/gnb.yml} # Path to your desired config file
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-192.168.70.5}
          bind_addr: ${GNB_IP:-192.168.70.10}
      metrics:
        autostart_stdout_metrics: true
        enable_json: true
      remote_control:
        bind_addr: 0.0.0.0
        enabled: true
  ue_config.conf:
    file: ${UE_CONFIG_PATH:-./configs/ue.conf} # Path to your desired config file

volumes:
  gnb-storage:
  ue-storage:

networks:
  ran:
  ipam:

```

```
    driver: default
    config:
      - subnet: 192.168.70.0/24
metrics:
  ipam:
    driver: default
    config:
      - subnet: 172.19.1.0/24
```